



PSOC™ Edge: Encryption

Customer training workshop (CTW)

January 2026



Prerequisites

Training prerequisites

- Introduction to PSOC™ Edge and ModusToolbox™
- For an introduction to PSOC™ MCUs, including a getting started guide to ModusToolbox™, visit: <https://www.infineon.com/product-information/psocdeveloper>
- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, visit: [PSOC™ Edge E84 Training Collection](#)

Development tools

- ModusToolbox™ v3.6
- Eclipse IDE for ModusToolbox™ v2025.4.0
- Edge Protect Security Suite v1.6
- ModusToolbox™ Programming Tools v1.5.0
- KIT_PSE84_EVAL_EPC2 BSP v1.0.0
- Terminal Emulator
- KIT_PSE84_EVAL

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Introduction

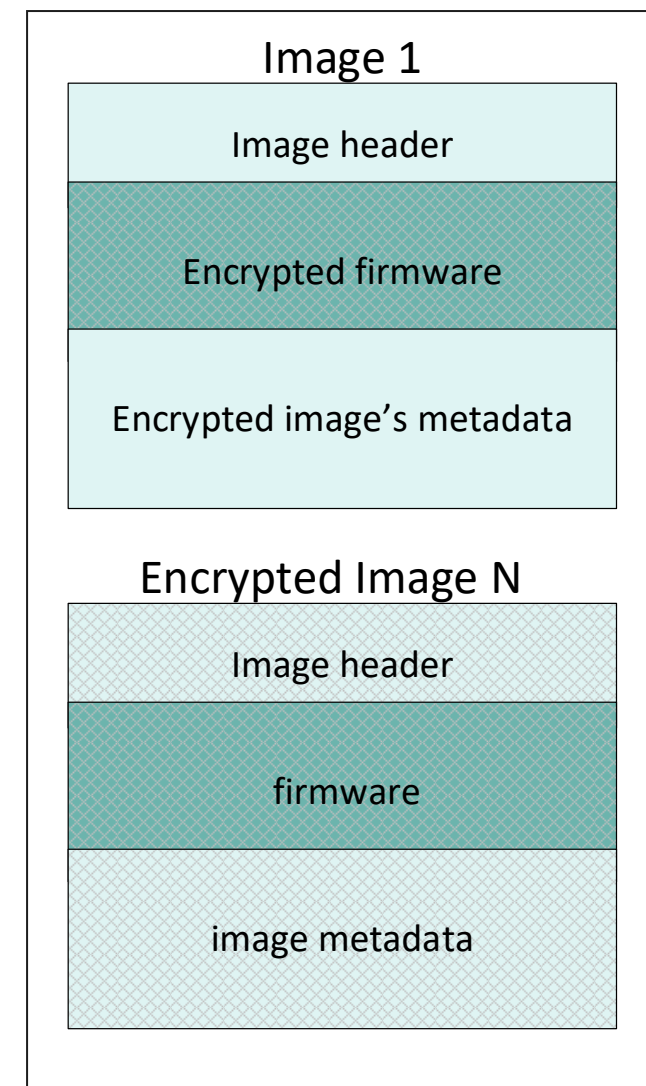
- Firmware image can be encrypted and programmed into device memory
- Edge Protect Bootloader (EPB) supports booting encrypted firmware only when running from RRAM
- Encrypted user images must reside in external flash and either executed in SMIF XIP mode (for EPC 2) or loaded to SRAM for execution (for EPC 4)
- Edge Protect Bootloader supports encrypted application execution via:
 - Single key encryption (EPC 2)
 - Multi key encryption (EPC 2)
 - Secure encryption (EPC 4)

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Single-key XIP encryption: EPC 2

- All three application images use the same AES encryption key
- The first image contains the AES key and AES nonce (part of the initial vector), packed in an HKDF block and stored in a TLV.
- For the first image, only the payload (application code) is encrypted; header and TLVs remain in plain text
- The second and subsequent images are fully encrypted (including header and TLVs) using the same AES key and nonce as the first image



Single-key XIP encryption: EPC 2 (contd.)

- An ECC key pair (Key-Encryption-Key or KEK) is used to encrypt the encryption key; the private portion of this KEK must be stored at a location known to the EPB. The default location is in the RRAM (0x3203A000)
- Supported with only overwrite upgrade mode
- See [Encrypted Boot](#) code example for enabling single-key encryption

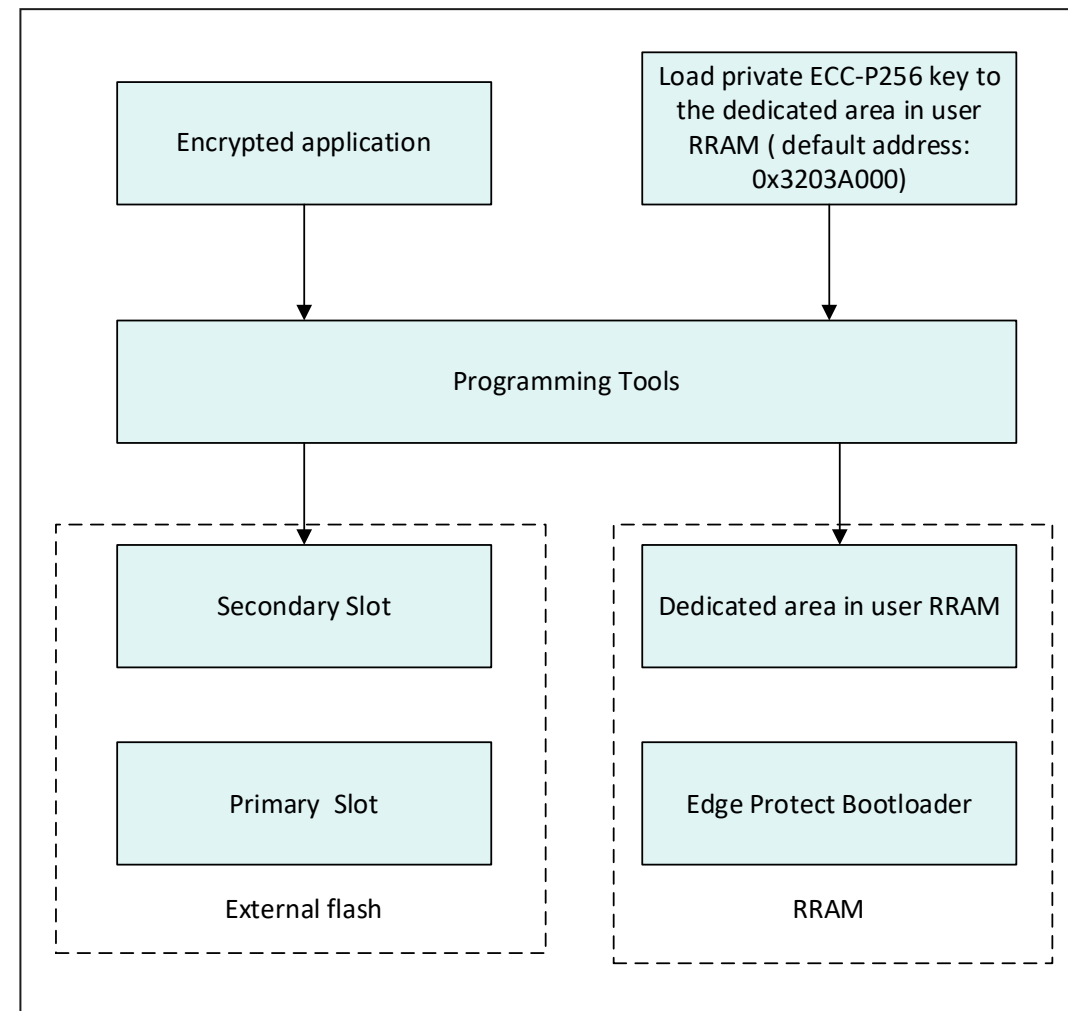
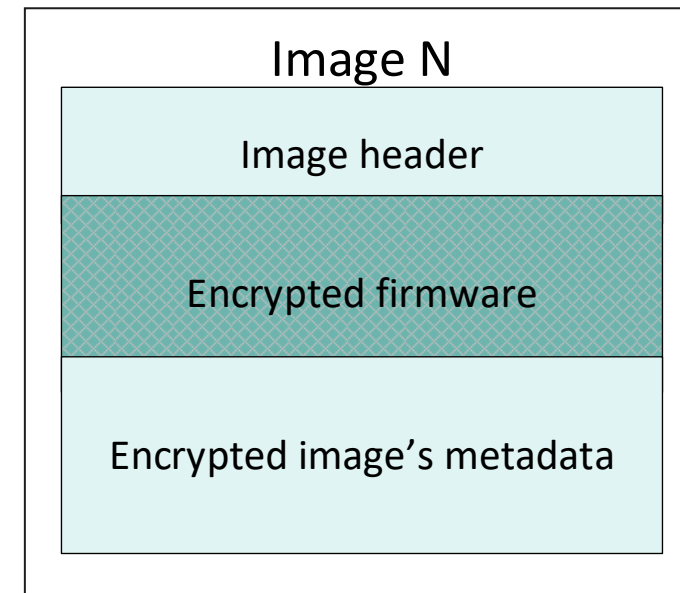


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Multi-key XIP encryption: EPC 2

- Each image always contains its own AES key and AES nonce (part of initial vector) packed to HKDF block and stored in one of TLV in the image
- Up to 4 different encrypted areas and different AES keys are supported
- Only the image payload (application code part) is encrypted; the header and image TLV are stored as plain data
- Multi-key XIP encryption has strict requirements for encryption region size and start address. The slot size of the application must be perfect power-of-two number. The start address of the application image must be a multiple of its slot size i.e., $\text{start_address} = N * \text{size}$



Multi-key XIP encryption: EPC 2 (contd.)

- An ECC key pair (Key-Encryption-Key) is used to encrypt the encryption key; the private portion of this Key-Encryption-Key must be stored at a location known to the EPB. The default location is in the RRAM (0x3203A000)
- Supported with swap and overwrite upgrade mode
- See [Edge Protect Bootloader](#) Application Note for more details

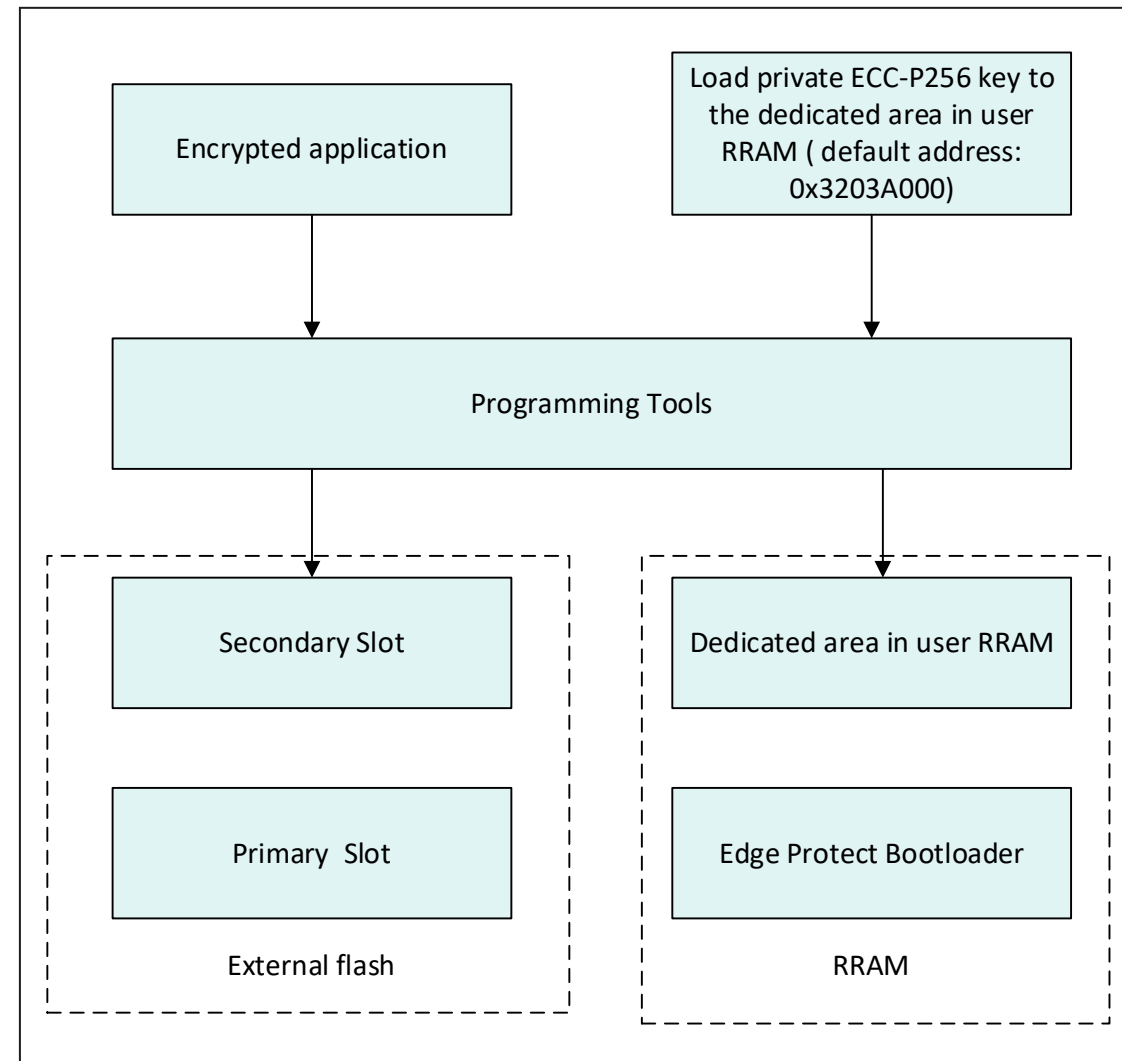


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Secure encryption: EPC 4

- On EPC 4 devices, a software-key based encryption mechanism is supported, compliant with PSA L4 certification
- For this use case, the private portion of the KEK must be provisioned into the device using the “user keys” section in the OEM policy file
- This method uses the AES CMAC algorithm for encryption
- This flow can only be used when the images are loaded into SRAM and executed from there
- See [Edge Protect Bootloader](#) Application Note for more details

Sample user keys policy for provisioning the KEK

```
{
  "kid": {
    "description": "The ID of the key in the PSA vault. The value between '0x40000000'
and '0x7FFFEFFF'",
    "value": "key_id"
  },
  "alg": {
    "description": "Key algorithm",
    "applicable_conf": "ECDSA, RSA, AES-CTR, AES-CBC, AES-ECB, ECDH, HKDF_ALG_SHA_256,
NONE",
    "value": "ECDH"
  },
  "use": {
    "description": "Key usage flags are encoded in a bitmask (usage example SIGN|
ENCRYPT)",
    "applicable_conf": "EXPORT, COPY, CACHE, ENCRYPT, DECRYPT, SIGN_MESSAGE,
VERIFY_MESSAGE, SIGN_HASH, VERIFY_HASH, DERIVE, VERIFY_DERIVATION",
    "value": "DERIVE"
  },
  "key": {
    "description": "Path to the key file in PEM, DER, or JWK format",
    "value": "../keys/enc-ec256-priv.pem"
  },
  "owner": {
    "description": "Key owner identifier. The value between '-2147483648' and
'2147483647'",
    "value": 0
  }
}
```

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Lab: Encryption support with EPB

Boot encrypted image with Edge Protect Bootloader

Revision history

Document revision	Date	Description of change
**	2026-01-14	Initial release.

